

Corporate Finance
PGE M1 Fall 2026

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Session 3: Investment Decision Rules

Housekeeping

About me

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Review: NPV Rule

Definition. $NPV = PV \text{ of benefits} - PV \text{ of costs}$

NPV Rule. Invest if $NPV > 0$

- ▷ If projects are mutually exclusive (you can only pick one or the other), choose the project with the higher NPV.
- ▷ If there are no constraints, invest in all projects with a positive NPV.

When can constraints matter?

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-10
1	120	21
NPV	9.09	9.09

When can constraints matter?

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-100
1	120	0
2	0	0
3	0	0
4	0	0
5	0	175.69
NPV	9.09	9.09

Internal Rate of Return

Definition. The **internal rate of return** (IRR) is the discount rate at which the NPV is equal to zero.

Normally, we compute the NPV as

$$\text{NPV} = -I + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3} + \dots$$

To compute the IRR, we set the NPV to zero and solve for the IRR

$$0 = -I + \frac{C_1}{1+\text{IRR}} + \frac{C_2}{(1+\text{IRR})^2} + \frac{C_3}{(1+\text{IRR})^3} + \dots$$

(using Excel)

The IRR Rule

The IRR is often interpreted as the average rate of return on an investment.

Example. Suppose you have an investment project that requires an initial investment of 100 but delivers 115 in one year. Assume $r = 10\%$.

- a. Compute the NPV of this project
- b. Compute the IRR



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The IRR Rule

The IRR Rule. Invest if the project has a IRR greater than the company's cost of capital k .

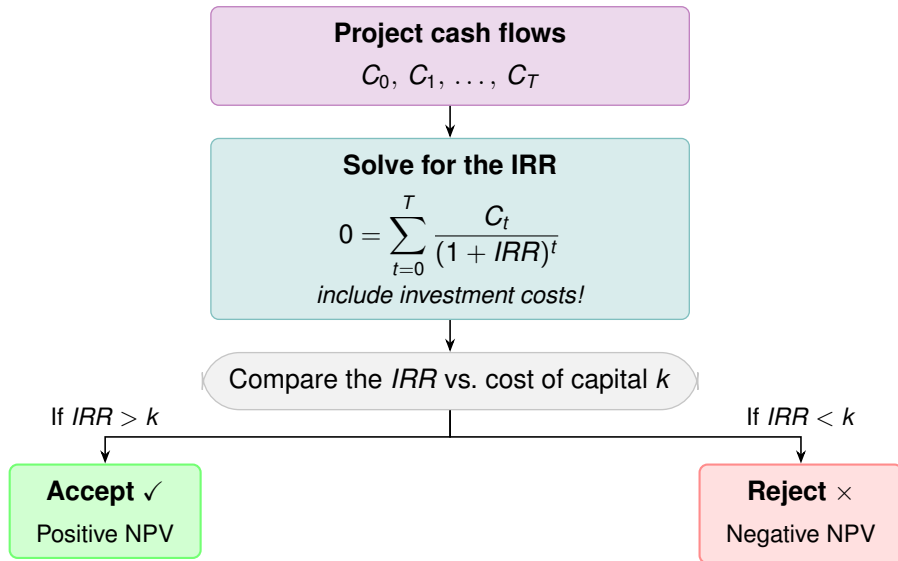
- Intuition: interpret the IRR as the average rate of return on a project. If the company's cost of capital exceeds the IRR, this implies the project's return is below the corporation's cost of financing / required rate of return.



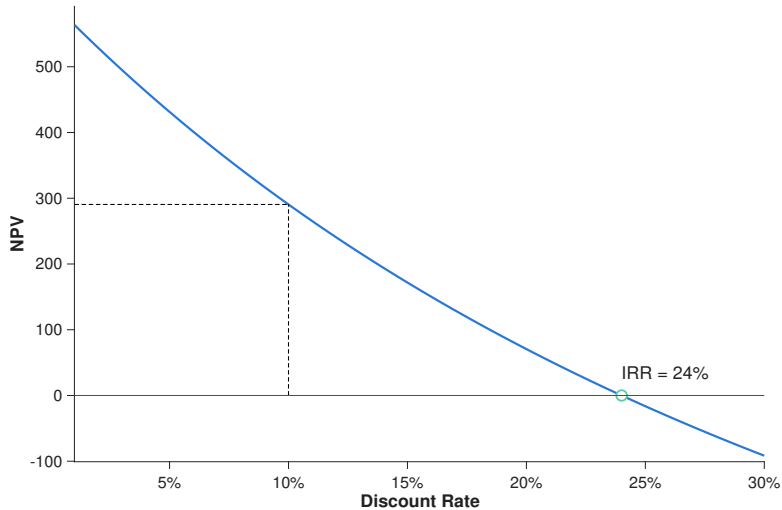
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The IRR Rule



NPV as a function of the IRR



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Pitfalls of IRR: Scale

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-10
1	120	12
NPV	9.09	.91
IRR	20%	20%

Both projects have the same IRR, but project A has a higher NPV.
IRR ignores **scale**.

Pitfalls of IRR: Timing

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-100
1	175	50
2	0	75
3	0	100
NPV	59.1	82.57
IRR	75%	47%

Project B has a higher NPV but Project A has the higher IRR.
IRR is sensitive to the **timing** of cash flows.

Pitfalls of IRR: Timing



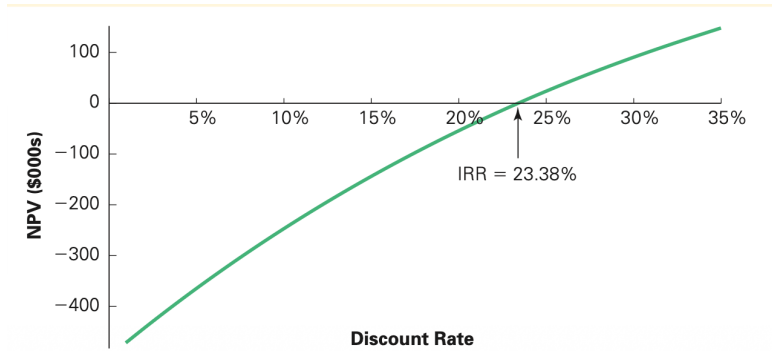
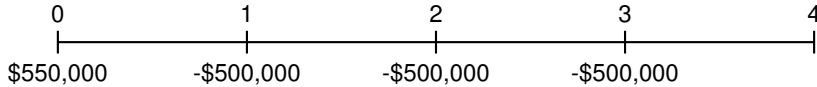
“Our equity strategy managed \$189 billion of AUM as of December 31, 2025 ... generating 39% gross IRR and a 24% net IRR on a compound annual basis.”

Apollo Global Management 2025 10-K

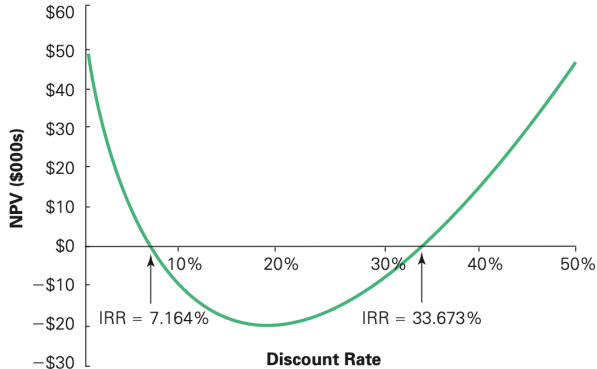
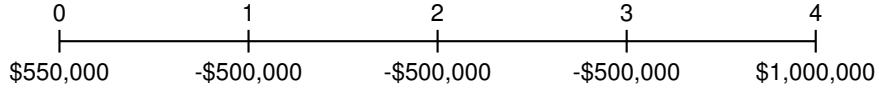
- ▷ That's \$28 bn in performance fees!

The nature of PE can generate high IRR by front-loading cash flows via recapitalizations through debt or cash.

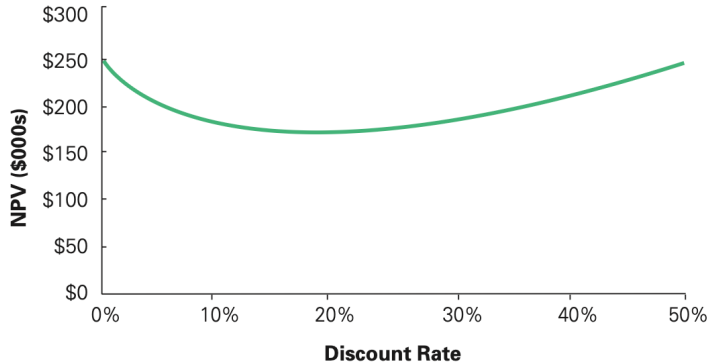
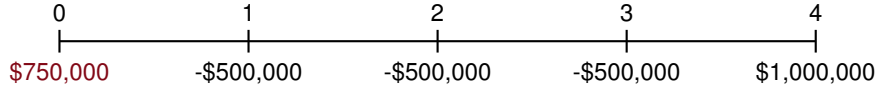
When the IRR breaks down: Delayed Investment



When the IRR breaks down: Multiple IRRs



When the IRR breaks down: Nonexistent IRR



Investment Decision Rules in Practice



“To summarize, MetLife’s deployed roughly \$3.7 billion of capital to support the origination of new business in 2022. This capital was put to work at an average internal rate of return of 17%, with an expected payback period of approximately 6 years. The value of new business generated in 2022 is around \$2.3 billion which represents the net present value of distributable cash flows in excess of the hurdle rate.”

MetLife Earnings Call transcript, 2023

Investment Decision Rules in Practice

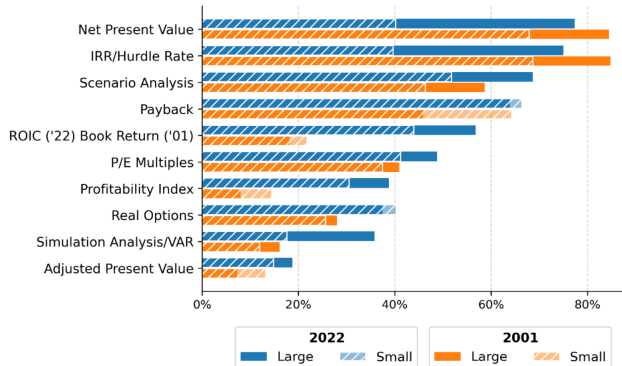


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Figure 2. Capital Budgeting Techniques

This figure displays CFO responses to the following question about the capital budgeting methods their companies use to make investment decisions: *How frequently does your firm use the following techniques when deciding which projects or acquisitions to pursue?* {0 = Never, 1, 2, 3, 4 = Always}. The percentage of firms that answer 3 or 4 are shown. The blue bars display results for the 2022 Duke CFO survey (March 2020 wave); the orange bars display results from Graham and Harvey (2001). Within each blue and orange bar, the solid portion displays responses for large firms (revenue above \$1 billion), and the white, crosshatched portion displays responses for small firms (revenue below \$1 billion). For example, 77% of large firms (40% of small firms) say that Net Present Value was an important or very important decision rule in 2022.



Exercise 1

You are considering opening a new plant. The plant will cost \$100 million upfront. After that, it is expected to produce profits of \$30 million at the end of every year. The cash flows are expected to last forever.

- a. Calculate the NPV of this investment opportunity if your cost of capital is 8%.
Should you make the investment?

NPV = 275m



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Exercise 1

You are considering opening a new plant. The plant will cost \$100 million upfront. After that, it is expected to produce profits of \$30 million at the end of every year. The cash flows are expected to last forever.

- b. Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged.

IRR = 30%



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Exercise 2

Bill Clinton reportedly was paid \$10 million to write his book *My Way*. The book took three years to write. In the time he spent writing, Clinton could have been paid to make speeches. Given his popularity, assume that he could earn \$8 million per year (paid at the end of the year) speaking instead of writing. Assume his cost of capital is 10% per year.

- a. What is the NPV of agreeing to write the book (ignoring any royalty payments)?
 $NPV = -9.895m$



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Exercise 2

Bill Clinton reportedly was paid \$10 million to write his book *My Way*. The book took three years to write. In the time he spent writing, Clinton could have been paid to make speeches. Given his popularity, assume that he could earn \$8 million per year (paid at the end of the year) speaking instead of writing. Assume his cost of capital is 10% per year.

- b. How many IRRs are there? Does the IRR rule give the right answer in this case?
IRR = 60.7%



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Exercise 2: continued

Now assume that, once the book is finished, it is expected to generate royalties of \$5 million in the first year (paid at the end of the year) and these royalties are expected to decrease at a rate of 30% per year in perpetuity.

- c. What is the NPV of the book with the royalty payments?

$$\text{NPV} \approx -.5\text{m}$$



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Exercise 2: continued

Now assume that, once the book is finished, it is expected to generate royalties of \$5 million in the first year (paid at the end of the year) and these royalties are expected to decrease at a rate of 30% per year in perpetuity.

- d. How many IRRs are there? Does the IRR rule give the right answer in this case?
IRR = 7.16% and 41.57%



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Exercise 4

You are deciding between two mutually exclusive investment opportunities. Both require the same initial investment of \$10 million. Investment A will generate \$2 million per year (starting at the end of the first year) in perpetuity. Investment B will generate \$1.5 million at the end of the first year and its revenues will grow at 2% per year for every year after that.

- a. Which investment has the higher IRR?

$$IRR^A = 20\% \text{ and } IRR^B = 17\%$$



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Exercise 4

You are deciding between two mutually exclusive investment opportunities. Both require the same initial investment of \$10 million. Investment A will generate \$2 million per year (starting at the end of the first year) in perpetuity. Investment B will generate \$1.5 million at the end of the first year and its revenues will grow at 2% per year for every year after that.

- b. Which investment has the higher NPV when the cost of capital is 7%?

$$NPV^A = 18.57\text{m and } NPV^B = 20\text{m}$$

Exercise 4

You are deciding between two mutually exclusive investment opportunities. Both require the same initial investment of \$10 million. Investment A will generate \$2 million per year (starting at the end of the first year) in perpetuity. Investment B will generate \$1.5 million at the end of the first year and its revenues will grow at 2% per year for every year after that.

- c. Use the incremental IRR rule to correctly choose between the investments. At what cost of capital would your decision change?

Incremental IRR = 8%

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Effective Annual Rate

Suppose a bank offers you .001% interest compounded every 6 hours. Is this a lot? A little? As a means of standardization, the **effective annual rate (EAR)** gives you the amount of interest that will be earned at the end of one year.



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Quoting Rates

In general, the standard way of quoting rates is

$r\%$ annually, compounded m times a year

such that you earn

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^{m \times T}$$

where compounded

- ▷ daily: $m = 365$
- ▷ monthly: $m = 12$
- ▷ quarterly: $m = 4$
- ▷ annually: $m = 1$

Quoting Rates

In general, the standard way of quoting rates is

$r\%$ annually, compounded m times a year

such that you earn

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^{m \times T}$$

and

- ▷ r : the nominal rate
- ▷ $\frac{r}{m}$: the proportional rate

Quoting Rates

“Suppose a bank offers you .001% interest compounded every 6 hours”

≠

“Suppose a bank offers you .001% interest **annually**, compounded every 6 hours”

(The nominal rate in the example is 1.46%)

Effective Annual Rate

We want

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^{m \times T} = \text{Deposit} \times (1 + \text{EAR})^T$$

Exercise 5

Compute the effective annual rate (EAR) for a one-year deposit:

- ▷ 10% compounded monthly (10.47%)
- ▷ 10% compounded annually (10%)
- ▷ 9% compounded daily (9.42%)